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Thermochemical Re-Pressurization to Mitigate Fracture Hits and Enhance Productivity in Tight Formation Infill Drillings

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Abstract

Infill drilling in unconventional reservoirs often leads to stress redistribution and pressure depletion around the parent well. This well interference results in fracture hits between parent and child wells, reducing stimulation effectiveness and production from both wells. This work presents an in-situ thermochemical re-pressurization treatment to increase pressure in the depleted region near the parent well prior to child well treatment, with the objective of protecting the parent well while improving reservoir access for the child well. A reactive fluid system is injected into the parent well to trigger a controlled exothermic reaction that generates in-situ heat and pressure, increasing the local state of stress to recharge the pressure in the well drainage zone. The treatment is conducted in multiple stages in the parent well, while the child well remains shut-in. Once adequate re-pressurization levels are achieved, the child well can be stimulated. The elevated stress field near the parent well will redirect fractures in the child well(s) towards less-depleted and undisturbed regions of the reservoir, where higher pore pressure and intact rock support more effective fracture propagation. In addition to evaluating critical engineering controls required for downhole implementation, candidate chemical systems are evaluated, and competing thermodynamic reaction pathways are analyzed. A process simulation methodology was developed to model the heat and mass balance of the primary candidate reaction, Ammonium Chloride and Sodium Nitrite, under representative unconventional reservoir conditions. The simulation results confirm that the system can generate considerable heat and a gas phase composed of Nitrogen and steam, resulting in the thermal expansion of fluids and the formation, thus re-pressurizing the parent well. This approach makes it possible to avoid fracture overlaps with existing stimulated volumes and facilitates the development of new drainage pathways, which in turn improves overall stimulation coverage and reservoir contact efficiency.

Keywords: Exothermic Thermochemical Pressurization, Infill Drilling, Frac hit, Fracture Interference, Parent-Child Well Interaction, Enhanced Productivity

Introduction

Horizontal drilling and multi-stage hydraulic fracturing are two of the leading enablers for the development and success of unconventional resources. However, unconventional plays tend to have a low primary recovery factor, often less than 10% for tight oil reservoirs (Alfarge et al., 2017; Alvarez & Schechter, 2016). To maximize the recovery from these low-permeability formations, infill drilling has become standard practice (Roussel et al., 2013). Unlike multi-lateral wells, new wells, often called child wells, are drilled adjacent to existing wells, which are referred to as the parent wells (King et al., 2017).

Figure 1 shows a typical configuration involving a parent well (i.e., the first well drilled in the reservoir) and two child wells on its either side. While this configuration enhances reservoir coverage, it often results in parent-child well interference. Parent well production depletes the drainage area, reducing pore pressure and affecting the effective stress (defined as the minimum horizontal stress minus pore pressure). As a result, the formation stress in the depleted zone drops significantly (Safari et al., 2017). When a child well is fractured nearby, the induced fracture tends to propagate preferentially toward this lower-stress, depleted zone, rather than toward undrained regions where pore pressure remains high (Safari & Ghassemi, 2016).

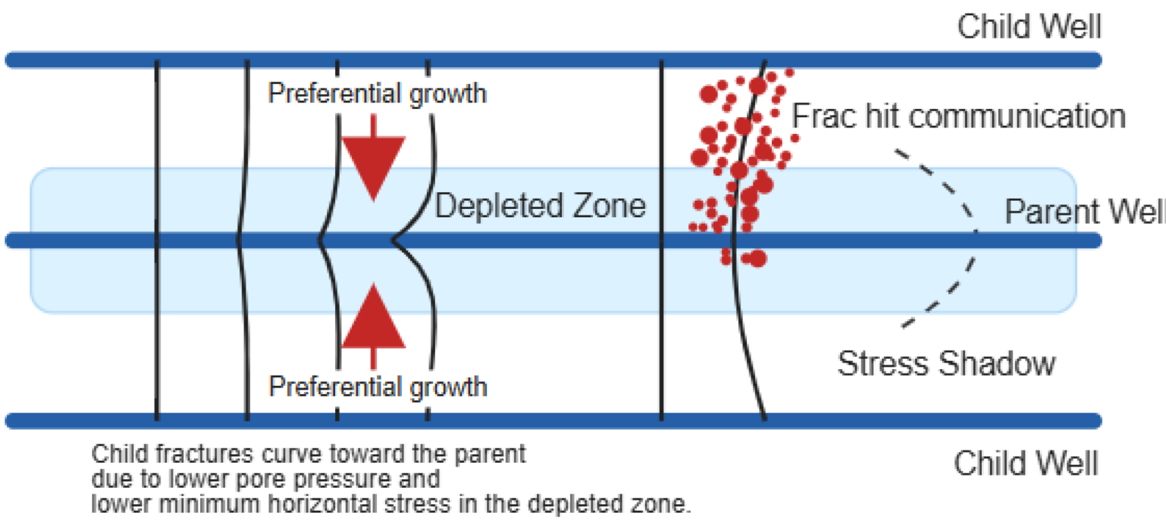

Figure 1—Schematic of a Parent/Child well. Child well fractures grow towards the depletion zone of the parent well.

Field studies support this behavior as well. In the Sichuan Basin, production from a parent well led to pore-pressure drops of up to 20 MPa in the surrounding reservoir over approximately three years, inducing a shifting stress regime that caused fracture interference during child-well fracturing (Tang et al., 2025). This aligns with observed phenomena of Frac Interference, Fracture-Driven Interactions (FDIs) or "frac hits" and complexities in multi-well frac designs due to stress and pressure interactions.

Pore pressure depletion propagates widely into adjacent shale overburdens (Ricard et al., 2012). Pressure diffusion studies have shown that reservoir pore pressure decreases can extend tens of meters or more into shale (Duda et al., 2023). Together, these findings underscore how parent-well depletion alters both pressure and stress landscapes, potentially directing fracture propagation towards areas of stress sink. Quantifying the magnitude of these stress drops and their spatial reach is therefore essential in designing safe and efficient infill and fracture treatments.

The stimulation energy meant for the child well is wasted in pre-existing fracture networks, thus leading to poor reservoir contact and suboptimal production (Guo et al., 2021; King et al., 2017; Rainbolt & Esco, 2018). The parent well integrity can also be damaged by the injection of fracturing fluids and proppant as they infiltrate the existing fractures in the parent well, causing mechanical or chemical damage (Jacobs, 2021; Safari et al., 2017; Safari & Ghassemi, 2016), and pose mechanical stress on casing and formation, which can lead to a sharp decline in its production rate (Jacobs, 2021). Additionally, asymmetrical stress redistribution and pressure fluctuations during frac hits may induce microfractures or stress damage, which can compromise well structural integrity (Whitfield et al., 2018) or fracture conductivity. The productivity of the infill or child well can also be compromised due to its drainage from the same depleted reservoir segment and suboptimal reservoir contact, leading to a reduction in its overall recovery potential.

To mitigate the risk of frac-hits, measures such as re-pressurizing the parent well have been proposed, aiming to restore reservoir pressure and stress conditions, thereby minimizing undesired fracture growth and communication between wells (Haghighat & Ewert, 2022). Pressure build-up around the wellbore was evaluated under various conditions, including parent well shut-in, gas injection, and water injection (pre-loading). Among these, water injection produced the most significant pressure increases in the vicinity of the wellbore. This process is often called "pre-loading" and can involve pumping volumes as small as 500 to 1,000 bbls (small pre-loads, Whitfield et al., 2018) to practical large volumes (18,000 to 45,000 bbls, Whitfield et al., 2018) and theoretical maximums of up to 300,000 barrels of water into the parent well prior to stimulating the child well (Wu et al., 2023) which could potentially mitigate the risk of frac-hits by reducing the tendency for fracture propagation toward neighboring wells. However, it is important to note that unloading the injected water typically requires a period of several months, which may impact operational timelines and productivity (Garza et al., 2018; Haghighat & Ewert, 2022). Additionally, water blockage can damage the well productivity(Fakcharoenphol et al., 2014). We also need to consider the fact that the resulting pressure build-up may not be enough to avoid fracture hits in the first place, if the parent and child well placement is not favorable (Wu et al., 2023).

Re-pressurizing an offset well with natural gas or carbon dioxide (Jacobs, 2017) can also help avoid well damage caused by water injection, but it is an expensive alternative. Additional considerations need to be made for the volume of the fluid that is needed to re-pressurize the parent well, which can pose significant logistical and economic burdens for the project (Joslin et al., 2020; Wu et al., 2023; Zheng et al., 2021). A simple practice of shutting in the parent well before fracturing the child well may be enough to build up sufficient pressure and stress and limit the likelihood of fracture impacts (Jacobs, 2017). However, this practice is not encouraged as the number of shut-in days is uncertain, leading to financial loss due to production obstruction.

This paper introduces and evaluates the use of an in-situ, exothermic reaction through mixing of Ammonium Chloride (NH_4Cl) and Sodium Nitrite ($NaNO_2$) to create a localized zone of high pressure and high temperature around the parent well. The primary objective is to establish a robust temporary stress barrier that redirects child well hydraulic fractures into virgin, undepleted regions of the reservoir. This work presents the theoretical basis for this approach, evaluates the thermodynamics of candidate chemical systems, and provides a rigorous process simulation to validate the core principles of heat and gas generation under reservoir conditions.

Related Work

Chemical System and Reaction Pathway Analysis

Several candidate reagents and chemical systems were evaluated for their potential to achieve the desired pressure build-up and facilitate the formation of a stress barrier. The candidates considered included Sodium percarbonate, urea, Nitrites, Ammonium salts, hydrogen peroxide, polyhydroxy aldehydes, and chromium trioxide, among others.

However, a number of these options were promptly eliminated due to significant toxicity and environmental or health hazards—for example, chromium-based compounds were excluded due to their known carcinogenic and ecological risks. Other candidates were deemed impractical due to various factors, such as:

- the generation of reactive oxygen species,

- the risk of adverse reactions with reservoir fluids,
- economic limitations, or
- regulatory restrictions related to explosive handling and transport.

This screening process ensured that only safe, feasible, and cost-effective systems were retained for further evaluation in the context of subsurface chemical treatment.

The primary candidate system proposed by this study is the aqueous reaction between Ammonium Chloride (NH_4Cl) and Sodium Nitrite ($NaNO_2$), which results in high exothermicity, significant gas generation, and environmentally friendly and safe byproducts. The reaction releases a substantial amount of energy, which will result in enough heat to overcome stress shadowing and provide formation re-pressurization. Production of inert Nitrogen gas, along with vaporization of water (both the water used as a carrier and the water resulting from the reaction), results in the volumetric expansion necessary to increase pore pressure. The reaction products themselves, which include Nitrogen, water, and Sodium Chloride, are environmentally benign and pose minimal risk of formation damage. Additionally, the resulting Sodium Chloride can further stabilize the shale and prevent any swelling that is anticipated.

The reaction between Ammonium Chloride and Sodium Nitrite is the most commonly used reaction in the oil and gas industry for heat generation (Al-Taq, Aljawad, Alade, et al., 2023; Qian et al., 2019). The reaction between Ammonium ion (NH_4^+) and Nitrite ion (NO_2^-) ions in aqueous solution represents a highly exothermic redox process, releasing Nitrogen gas (N_2) and water. It is also one of the most commonly used reactions to produce Nitrogen in the laboratory, due to its ease of use and safety. The kinetics of the chemical reaction in aqueous solution have been studied as a function of pH, temperature, and reactant activities. At 25 °C the enthalpy change for the 1:1 reaction in aqueous solution is about −79.95 kcal mol^{-1}, which is −335 kJ mol^{-1}(Nguyen et al., 2003):

$$NH_4Cl(aq)+NaNO_2(aq)\rightarrow N_2(g)+2H_2O(l)+NaCl(aq) \quad \Delta H^0\approx -335 \ \ kJ/mol \tag{1}$$

Beyond heat generation, the reaction is tunable through pH and formulation, which has driven field use across several flow-assurance and stimulation tasks. One of the most promising industrial applications of this reaction is in remediating paraffin and asphaltene deposits in deep-water oil pipelines. The reaction between Ammonium Chloride and Sodium Nitrite is uniquely suited for this application due to its intense exothermicity, rapid kinetics, and ability to be fuse delayed, or triggered as needed. For wax and asphaltene control, pH-managed formulations have been designed specifically to fluidize low-melting deposits while maintaining process safety in wells and lines (Bispo et al., 2021). Field-scale and lab studies report paraffin remediation and integration with fracturing fluids; for example, an in-situ heat fracturing fluid based on $NH_4Cl/NaNO_2$ was formulated to reach about 50 °C and showed improved production in a five-well comparison (Mao et al., 2021).

Thermochemical fluids built on this same chemistry have also been used to remove emulsion blockages created by oil-based muds (Alade et al., 2020, 2021). Alade et al. (2020, 2021) reported raised temperature from about 100 °C to roughly 170 °C and pressure to about 1600 psi in controlled setups, with about 72% removal of water-in-oil emulsion after injecting one pore volume.

For EOR and near-wellbore energy enhancement, in-situ N_2-generation studies confirm how reactant ratio, temperature, and acid precursors affect gas yield and energy release. A kinetic study that developed the gas-generation rate identified a practical 1:1 molar ratio for $NH_4Cl:NaNO_2$ and quantified the dependence on H^+ and temperature, with results aimed at oilfield deployment (Qian et al., 2019).

Due to its heat generation and controllability, this reaction forms the basis of an emerging class of systems known as fused chemical reactions, whose onset can be delayed or activated through pH, encapsulation, or thermal conditions. These reactions are increasingly used across sectors, including pharmaceutical drug delivery (Shahzad et al., 2015), smart fertilizers (Feng et al., 2015), paraffin and asphaltene deposition in oil pipelines (Mao et al., 2021), and subsurface flow assurance systems in petroleum engineering (Al-Taq, Aljawad, Alade, et al., 2023; Bispo et al., 2021).

The redox reaction between total Ammonium, which is the sum of free ammonia and its conjugate acid ($NH_3 + NH_4^+$), and total Nitrite (i.e., the sum of Nitrite and its conjugate acid ($NO_2^- + HNO_2$)), is very slow at neutral or basic conditions(Nguyen et al., 2003). However, this reaction accelerates several-fold in acidic media(Nguyen et al., 2003).

The rate was found to be first order with respect to total Ammonium concentration and second order with respect to Nitrite. At low concentrations of Nitrite ion and low pH values, the reaction follows second-order kinetics where the reaction between nitrosyl ion (NO+) and molecular ammonia (NH3) determines the rate of the reaction (Dusenbury & Powell, 1951). Reaction rates increase substantially as the pH level drops. For example, the reaction rate at pH 5 is approximately 138 times faster than at pH 7, and the rate at pH 3 is about 4,000 times faster than at pH 7 (Nguyen et al., 2003). This strong acid effect reflects shifts in acid–base and nitrosating equilibria that raise the concentrations of the reactive HNO_2 and N_2O_3, rather than a change in the elementary pathway (Nguyen et al., 2003).

The reaction is also thermally activated, so higher temperatures increase the rate, a dependence that has been quantified for both acid-activated and heat-activated Nitrogen-generation systems used in downhole applications (Al-Taq, Aljawad, Alfakher, et al., 2023; Bispo et al., 2021; Nguyen et al., 2003; Qian et al., 2019). While the overall stoichiometry summarized in Equation (1) reflects the desired reaction pathway, these pH and temperature dependencies provide practical means for rate control in the wellbore (Nguyen et al., 2003).

While the application of this specific reaction for well stimulation is established, its use as a geomechanical tool to create a stress barrier for frac hit mitigation is a direct extension that has not been explored before. Previous work on frac hit mitigation has focused almost exclusively on mechanical diversion techniques or large-scale fluid injection. Several researchers have explored the optimization of well spacing and fracture design to minimize interference (Bazan et al., 2010; Iino et al., 2018; Liang et al., 2019; Sen et al., 2018). Others have modeled the impact of re-pressurization using water or gas, confirming the potential of stress manipulation and highlighting the associated operational challenges and risks (Gala et al., 2018). This study bridges a gap by proposing a more targeted, efficient, and potentially less damaging method of stress alteration using proven thermochemical principles.

Methodology

A multi-step approach was taken to identify the ideal chemical system and understand potential reaction pathways, which play a crucial role in downhole reaction control.

Simulation Methodology

To validate the foundational concept, the reaction was modeled using the CHEMCAD 8.1.2 process simulator. The primary objective was to quantify the heat and gas generation under simulated reservoir conditions, and the development of a successful model required a systematic troubleshooting process.

In order to ensure that the most probable reaction pathway under representative reservoir conditions is modeled, our approach must be grounded in thermodynamics. For this reason, we employed Gibbs Free Energy (GFE), which provides the fundamental criterion for chemical equilibrium. In a system held at constant temperature and pressure, the spontaneous direction of reaction corresponds to a decrease in GFE, with equilibrium reached at its minimum value (Atkins, 2010). Accordingly, a Gibbs Free Energy minimization framework is the best method for evaluating all possible product species and predicting the final chemical composition that corresponds to the lowest total energy of the system and ensures identification of the most thermodynamically probable outcome of the reaction.

In order to simulate bottom-hole conditions characteristic of the typical shales, a range of values from different sources was gathered, Table 1. If values were reported in gradient form (e.g., BHP for Eagle Ford was reported as 0.71 psi/ft by Bozeman et al. (2023)), they were adjusted to account for depth.

Table 1—Representative Bottom-hole Conditions for Different Shale Plays in the US

Case No.	Shale play (basin)	Depth (ft)	BHP (psi)	BHT (°F / °C)	Source
1	Wolfcamp (Permian, Delaware Basin)	10,000 ft	4,500 psi	150 °F (66 °C)	(Ramiro-Ramirez et al., 2024)
2	Bakken (Williston Basin)	10,000 ft	5,300 psi	220°F (104 °C)	(Alvarez & Schechter, 2016; Sonnenberg & Pramudito, 2009)
3	Haynesville (East TX–NW LA)	12,000 ft	8,800 psi	380 °F (193 °C)	(Arnone et al., 2022; Hammes & Frébourg, 2012)
4	Eagle Ford (South Texas)	10,000 ft	7,100 psi	235 °F (113 °C)	(Bozeman et al., 2023)

We utilized the Gibbs Reactor module in CHEMCAD to account for all potential reactions and competing pathways without requiring predefined reaction kinetics or any time dependency, which makes it an ideal tool for equilibrium-based analysis. The simulation was performed under constant pressure and adiabatic conditions, meaning no heat loss to the surroundings was considered to quantify the maximum potential temperature rise. The reactor pressure was set to the initial reservoir pressure and temperature listed in Table 1 to replicate reservoir conditions.

Table 2 shows the components considered for the reaction. The Gibbs reactor was configured to be adiabatic to quantify the maximum potential temperature increase, and the system pressure was held constant. We also considered several reactions that can take place, as well as the electrolytic activities that can occur in the medium. Due to the presence of dissolved salts, the Electrolyte thermodynamic package was utilized for all property calculations. Table 3 lists the electrolytic reactions that were considered in the chemical modeling. Along with the inlet materials and materials expected to be observed in the outlet, possible intermediate materials (e.g., Ammonium Nitrate) were also considered.

Table 2—Components Considered for the Reaction

Name	Chemical Formula	Stream
Ammonium Chloride	NH_4Cl	Inlet
Sodium Nitrite	$NaNO_2$	Inlet
Nitrogen	N_2	Outlet
Sodium Chloride	$NaCl$	Outlet
Water	H_2O	Inlet, Outlet
Ammonium Nitrate	NH_4NO_3	Intermediate, Outlet

Table 3—Electrolytic Reactions Considered

$NaCl \leftrightharpoons Na^{+} + Cl^{-}$	(2)
$NaNO_2 \rightleftharpoons Na^{+} + NO_2^{-}$	(3)
$NH_4Cl \rightleftharpoons NH_4^{+} + Cl^{-}$	(4)
$H_2O \rightleftharpoons H^{+} + OH^{-}$	(5)
$NH_4NO_3 \rightleftharpoons NH_4^{+} + NO_3^{-}$	(6)

Figure 2 shows a schematic of the model created in CHEMCAD. Two separate streams of Ammonium Chloride and Sodium Nitrite enter the Gibbs reactor at reservoir temperature and pressure, where they are mixed with each other and react in an adiabatic fashion. The products are subsequently taken into a vertical flash vessel to separate the gas stream from the liquid stream. It should be noted that this separation is only meant to be used for calculation purposes, and does not actually happen in the reservoir.

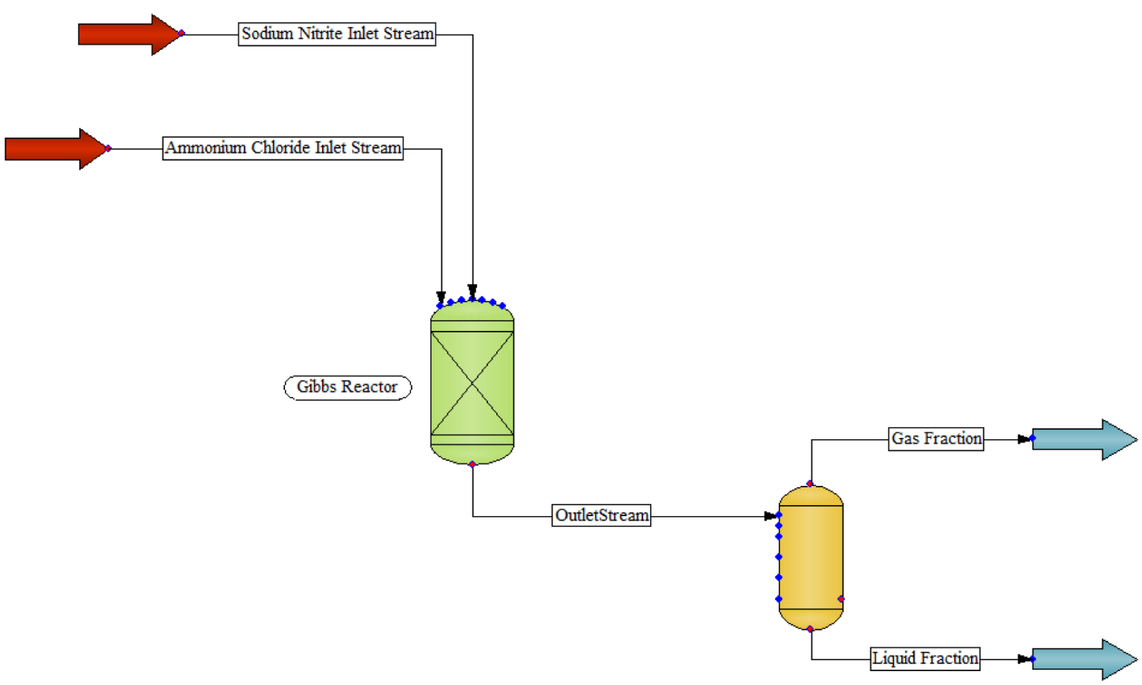

Figure 2—CHEMCAD Simulation Model Composed of a Gibbs Reactor and a Flash Vessel to Model Injection Reaction and the Resulting Products

The reaction extent was 100% fractional conversion of the limiting reactant, Sodium Nitrite. Sodium Chloride was set as the potential solid that would form as a product of this reaction. Since we are considering all reagents in Equation (1) to be in aqueous phase, we included a significant volume of water to be fed to the reactor as one of the feeds, while additional water molecules are considered to be added as byproducts, expected in the outlet stream.

In order to estimate reaction outcomes under operational and practical conditions, we will use observations made by Whitfield et al. (2018), where injection rates of 12-15 bbl/min, along with large volumes (18,000 – 45,000 bbls), were considered effective in preventing frac-hits but were not economically favorable. As a result, we will try to stay below a 6 bbl/min injection rate and maintain injected volumes to less than 9,000 bbls. We assume separate aqueous streams for NH_4Cl and $NaNO_2$, mixed only at or near the wellbore.

The effectiveness of the treatment is highly dependent on the concentration of the reagents. To maximize the re-pressurization effect, the injected fluid must be as concentrated as possible. An optimal design will use the minimum amount of water that is practically required to fully dissolve the reactants and allow for safe pumping. The calculation is predicated on determining the minimum molar ratio of water to solute required to form a saturated solution, which represents the theoretical limit for the most concentrated injectable fluid. Assuming a surface temperature of 68 °F (20 °C), we can extract the solubility and density of the total solution provided by Sigma Aldrich (2025). Table 4 shows the solute properties at 68 °F (20 °C).

Table 4—Solute Properties at 68 °F (20 °C). Solubility and Density Values Extracted from Sigma Aldrich (2025)

Solute	Weight (kg/kmol)	Properties at 68 °F (20 °C)			
		Solubility (gr/100gr H_2O)	Weight Percentage (%)	Density of Total Solution (kg/L)	Solute Mass (kg/bbl)
Ammonium Chloride (NH_4Cl)	53.49	37.56	27.3	1.075	46.66
Sodium Nitrite ($NaNO_2$)	69.00	84.52	45.8	1.33	96.85

The weight percentage is determined by dividing the soluble weight of the solute by the total weight of the solution. For example, for NH_4Cl, the solubility at the given temperature is 37.56 grams per 100 grams of water; thus, the weight percentage of NH_4Cl in the final solution becomes 37.56/(100+37.56) or 27.3%. Likewise, the mass of solute in each barrel of solution is determined by converting one barrel of solution to its equivalent in liters (i.e., 158.987 *L/bbl*) and calculating the solution mass, then accounting for the solute mass by considering its weight percentage. As an example, the solution mass for each barrel of NH_4Cl becomes $1.075\frac{kg}{L}*158.987\frac{L}{bbl}=170.91\ \ kg$, and the resulting NH_4Cl solute mass will be $27.3*170.91\frac{kg}{bbl}=46.66\frac{kg}{bbl}$.

The resulting molar flow rates to be used in CHEMCAD are determined by determining the number of moles of the solute in each barrel of solution, and converting the injection rate (e.g., $6\frac{bbl}{min}$) on an hourly basis. As an example, the number of moles of NH_4Cl per bbl of solution can be calculated as 46.66 *kg* / 53.49 *kg/kmol* = 0.872 *kmol/bbl* . Assuming that we are injecting at $6\frac{bbl}{min}$, the flow rate of NH_4Cl will be $0.872\frac{kmol}{bbl}*2.2046\frac{lb}{kg}*6\frac{bbl}{min}*60\frac{min}{hr}=692\frac{lb.mol}{hr}$. A similar calculation for $NaNO_2$ would result in an injection rate of $1,114\frac{lb\cdot mol}{hr}$ of $NaNO_2$. In order to safeguard the operation and account for brief temperature dips, one can cap saturation concentration at 80%, which will result in $554\frac{lb\cdot mol}{hr}$ of NH_4Cl and $891\frac{lb\cdot mol}{hr}$ of $NaNO_2$, respectively.

We can also estimate the flow rate of water for each solution by considering that each barrel of solution is 158.987 L. Given the solution densities, Table 4, we can estimate the mass of each barrel of solution as its $density\ \left(\frac{kg}{L}\right)*158.987\ \left(\frac{L}{bbl}\right)$. This results in a total mass of water and solute of $1.075*158.987=\ \ 170.911\frac{\mathrm{kg}}{\mathrm{bbl}}$ for NH_4Cl solution, and $211.453\frac{kg}{bbl}$ for $NaNO_2$ solution. Subsequently, one can estimate the solute mass of water for the NH_4Cl solution by subtracting the mass of solute in each solution (i.e., $170.911\frac{kg}{bbl}-\ \ 46.659\frac{kg}{bbl}=124.252\frac{\mathrm{kg}}{\mathrm{bbl}}$ of water for the NH_4Cl solution). This means that each barrel of the NH_4Cl solution contains $124.252 \big/ 18.015 \;=\; 6.89596\frac{kmol}{bbl}$ of water. Since we are injecting at a rate of $6\frac{bbl}{min}$, we can expect $6.89596 \times 360 \;=\; 2,482.55\frac{kmol}{hr}$ or $5,472.2\frac{lb\cdot mole}{hr}$ of water. A similar calculation for the $NaNO_2$ solution results in $5,045.0\frac{lb\cdot mole}{hr}$ of water. The total water flow rate of water in the inlet stream is consequently equal to $5,472.2\frac{lb\cdot mole}{hr} + 5,045.0\frac{lb\cdot mole}{hr} = 10,517.2\frac{lb\cdot mole}{hr}$ of water. Table 6 lists the composition of the inlet stream.

Pressure Estimation

In order to determine the pressure increase that arises from injecting into isochoric (constant volume) conditions at the bottom-hole, as well as the temperature increase that arises from the chemical reactions (which are determined by ChemCAD), we modeled the pre-load as a coupled volume-balance and compliance problem at the near-wellbore scale. The initial effective storage volume V_0 was determined by considering the wellbore geometry as a vertical section, along with a lateral section. We used a 5.5 in production string (Bennett et al., 2020; Ridley et al., 2015)with a nominal internal diameter (d_{ID}) of 4.778 *in*. The capacity per 100 ft of casing (*Cap*) was calculated as:

$$Cap \;= \pi\left[\frac{\left(\frac{d_{ID}}{12}\right)^2}{4}\right] \times \frac{100}{5.615}bbl$$

The section volumes were subsequently calculated as $V_V = \frac{L_V}{100} \times \; Cap_V$ and $V_L = \frac{L_L}{100} \times \; Cap_L$, where V_V and V_L denote the vertical and lateral volumes, respectively. The total initial effective volume is thus calculated as $V_0 \;=\; V_V \;+\; V_L$.

Subsequent volume changes were determined by considering the total injected solution volumes (e.g., 9,000 bbls) along with the volume of the reaction products (water in liquid and vapor form, Nitrogen, and Sodium Chloride), as well as the change in volume due to thermal expansions that will be caused by the heat resulting from the reaction. We also considered a slow leak-off of injected fluids into the formation as a factor that reduces overall pressure build-up.

$$\Delta V_{Total} = \Delta V_{Inj} + \Delta V_{Gas}(T,P) + \Delta V_{Liq} + \Delta V_{Th} - \Delta V_{Leakoff}$$

Where injected volume (ΔV_{Inj}) is capped at the aforementioned solution volume of 9,000 bbls, and thermal expansion due to the release of reaction heat, ΔV_{Th}, is given as:

$$\Delta V_{Th} = \beta_T \Delta T.V_T$$

With β_T and V_T being effective bulk thermal expansivity (e.g., 3.0×10^{-4} °F^{-1}) and effective total volume, respectively. The change in volume due to gas generation at the reservoir temperature and pressure ($\Delta V_{Gas}(T,P)$) due to the expansion of Nitrogen and steam is determined as

$$\Delta V_{Gas}(T,P) = V_{std}\left(\frac{T}{T_{std}}\right)\left(\frac{P_{std}}{P}\right)\frac{1}{Z}$$

Once total volume change is determined, the total (or effective) system compressibility (C_t) is used to estimate the pressure change in the system.

$$\Delta P = \frac{\Delta V_{Total}}{C_t V_{Init}}$$

It should be noted that the water that is generated as a product of this reaction is considered to have the solid product, i.e., Sodium Chloride, already dissolved in it, and has a density of $65\frac{lb}{ft^3}$. As such, we can estimate that the generated water adds an incremental flow rate of $466.11 \frac{\text{lb}}{\text{hr}} * \frac{1}{65} \frac{ft^3}{lb} * \frac{1}{5.615} \frac{bbl}{ft^3} = 1.28 \frac{bbl}{hr}$, or $0.021 \frac{bbl}{min}$ to the injection stream.

Volume of the fluids that leaked off to the formation at each time step was estimated using a leak-off coefficient ($K_{Leakoff}$) which would be determined based on the formation permeability and any prior enhancements (stimulation or damage) to the wellbore, multiplied by the difference between the pressure in the previous timestep and the initial pressure, i.e., the pressure before pre-loading, not to be confused with the initial reservoir pressure. In this study, we have considered a leak-off coefficient of $2.0 \times 10^{-4} bbl/min/psi$:

$$V_{Leakoff} = K_{Leakoff} \times Max(P_{n-1} - P_i, 0)$$

Results and Discussion

Different reservoir conditions listed in Table 1 were studied to evaluate the impact of various temperature and pressure conditions on the reaction and the resulting product composition. Despite variations in the initial temperature and pressure, it was observed that the reaction followed the same path every time, and the only difference (in terms of the products and their composition) between different cases was the vapor phase fraction. The increase in temperature and pressure was consistently observed across different reservoirs, as shown in Figure 3. Figure 3 shows the evolution of bottom-hole pressure in case 2 (Haynesville) during injection and after shut-in. We can observe that the bottomhole pressure increases steadily as the injection process continues and reaction takes place, yet after shut-in, the pressure starts to taper off due to leak-off into the formation.

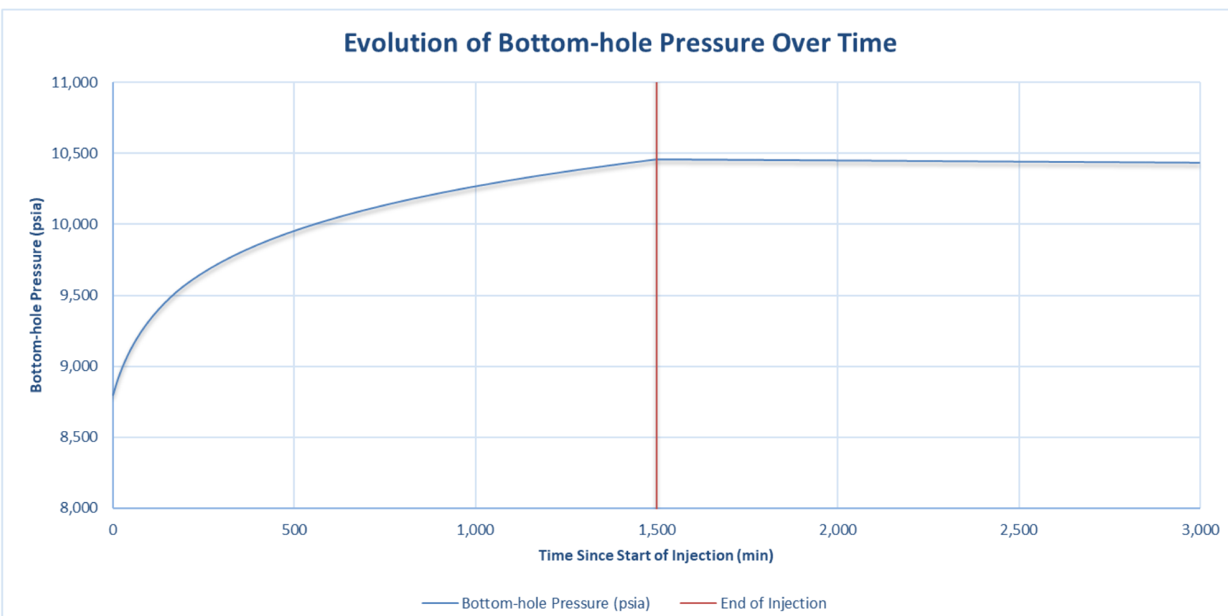

Figure 3—Evolution of Bottom-hole Pressure in Case 2 (Haynesville) During Injection and After Shut-In

Table 5—Initial and Final Conditions for Four Different Cases Modelled in this Study

	Initial Conditions		Final conditions		
	Temp. (F)	Press. (psi)	Temp. (F)	Press. (psi)	Vapor Frac. (%)
Case 1 (Wolfcamp)	150	4,500	310.3	6,197	1.21
Case 2 (Bakken)	220	5,300	374.8	6,993	1.08
Case 3 (Haynesville)	380	8,800	515.3	10,472	0.50
Case 4 (Eagle Ford)	235	7,100	385.7	8,775	0.77

It is critical to consider the influence of downhole conditions on this calculation as well. For a solid solute in a liquid solvent, the effect of pressure on solubility is generally negligible. However, the effect of temperature is significant. The solubility of Ammonium Chloride in water increases substantially with increasing temperature. Consequently, as the injected fluid heats up from surface conditions to the reservoir temperature, its capacity to hold the solute in solution will increase, which provides a safety margin to reduce the risk of premature crystallization of the salt within the wellbore or formation.

It should also be noted that water acts as a thermal load, absorbing the fixed energy released by the reaction. A larger water volume results in a lower peak temperature. Therefore, the effectiveness of the thermochemical treatment is critically dependent on the reagent concentration. To achieve the desired high-pressure, high-temperature barrier, the injected fluid should be as concentrated as is practically and safely feasible, minimizing excess water to maximize the in-situ thermal and volumetric expansion effects. It should be noted that even the diluted reagents still provide substantial amounts of energy to the formation, which in turn results in the expansion of the injected water and has the potential of providing additional pressure to the formation.

Table 6 shows the composition of the inlet and outlet streams. It should be noted that despite complete consumption of Ammonium Chloride, which leaves no trace of it in the outlet stream, there is about 19.87% of Sodium Nitrite left in the outlet stream, suggesting that optimization of its quantity is a critical design parameter that should be considered. Additionally, it is important to note that no trace of Ammonium Nitrate is detected in the outlet stream, suggesting that this unstable intermediate product is not present in the outlet stream; thus, it would not pose any issues for the process, as it is a harmful substance which could pose environmental and safety hazards.

Table 6—Composition of the Inlet and Outlet Streams

Name	**Chemical Formula**	**Flow Rate (lb.mole/hr)**	
		Inlet	**Outlet**
Ammonium Chloride	NH_4Cl	692	0
Sodium Nitrite	$NaNO_2$	1,114	221.4
Nitrogen	N_2	0	362.4
Sodium Chloride	NaCl	0	756.1
Water	H_2O	10,517.2	10,983.3
Ammonium Nitrate	NH_4NO_3	0	0

Downhole Reaction Control

A critical engineering challenge is to ensure the reaction proceeds only within the target reservoir zone, avoiding premature initiation in surface equipment or the wellbore. We envision three primary control strategies:

1. Our recommended procedure is sequential injection, where the two reactants are injected in separate slugs, isolated by an inert spacer fluid like brine. A slug of base can be injected ahead of the batch to neutralize any acid in the formation, which may result in the unintended start of the reaction. Ammonium Chloride in aqueous form is injected first, followed by a slug of brine, which helps push the reagent into the formation and away from the wellbore. Sodium Nitrite in aqueous form is injected next, followed by another slug of brine, which helps flush it out of the wellbore and brings it in contact with the Ammonium Chloride. Mixing and subsequent reaction are designed to occur within the pores and microfractures of the formation, far from the wellbore. In order to accelerate the reaction, the second reagent can be injected in a slightly acidic phase. This approach has the advantage of reduced risk of premature reaction, simpler deployment, broad applicability, and controlled placement; however, it suffers from an imprecise control over the mixing and potential leaving some of the zones untreated, depending on the heterogeneity of the formation.
2. The reaction rate is strongly dependent on pH. While the reaction rate is very slow in neutral or alkaline conditions, it accelerates significantly in an acidic environment. Therefore, a single fluid containing both reactants can be prepared at a high pH, rendering it stable for injection. The reaction can then be triggered downhole by a small, separate slug of acid or by relying on the natural conditions of the reservoir to gradually lower the pH. However, a significant consideration is the potential for unwanted side reactions with the formation itself. The primary reaction is, in many cases, catalyzed by acid. If an acid such as hydrochloric acid (HCl) is used to accelerate the reaction, it will readily react with any carbonate minerals, like calcite or dolomite, present in the shale formation. This secondary reaction would consume the catalyst, potentially slowing or halting the desired reaction, and would also alter the near-wellbore rock matrix. This approach has the advantage of a simpler injection schedule, enhanced stability during the injection, and flexibility of when the reaction is to be triggered. However, it poses a greater risk of unwanted side reactions and potential for formation damage. Additionally, existing pH levels of the formation can pose risks of premature triggering of the reaction.
3. Concentric coiled tubing can also be used, which utilizes specialized coiled tubing with an inner conduit and an outer annulus. The two reactant fluids are pumped down their respective channels and are only allowed to mix at the exit point of the tool, precisely at the target depth within the formation. This approach provides us with a very high level of precision both in terms of initiating the reaction, as well as placing it in the intended location, but it poses more financial burden to the project and can add the inherent risk of tool failure and operational limitations of coiled tubing.

Conclusions

This study successfully validated the scientific principles of using an in-situ thermochemical reaction for the parent well re-pressurization and pre-loading it for child well treatments while reducing the volume of water and horsepower needed to inject into the parent well by nearly half. The aqueous reaction of Ammonium Chloride and Sodium Nitrite was shown through process simulation to generate significant heat and subsequently temperature increases that result in the expansion of fluid volumes under typical reservoir conditions. We evaluated different reservoir temperature and pressure ranges and observed that the in-situ conditions did not have a significant effect on the reaction pathway, but resulted in different final temperature, pressure, and vapor fraction values.

An important consideration arises when the local pressure in the parent well exceeds the fracture gradient and secondary fractures are initiated. While these secondary fractures may enhance production from the parent well, the fact that these secondary fractures can affect child wells the same way that child well stimulations could affect the parent well should not be ignored. If the pressure is suspected to reach fracturing levels, modeling should be conducted to investigate the invasion of the drainage zone of the child well. In such cases, strategies such as pressure management, stimulation sequencing, and optimization of well placement may be considered to ensure that the child well maintains its independence and effective access to the reservoir.

In addition to reducing the required amount of water and horsepower needed to effectively pressurize the parent well, the proposed process is compatible with high-salinity, high-temperature environments and does not produce byproducts that can jeopardize the well integrity. Furthermore, this approach does not pose risks of formation damage (due to water blockage) or require costly infrastructure for CO_2 or gas injection; yet, the combined thermal and mechanical effects result in more effective stimulation and improved long-term recovery.

A sensitivity analysis needs to be established to identify an optimal range for the water-to-reagent ratio based on solubility limits, the resulting pressurization, and the economic advantages of using the proposed chemical reaction, vs. injecting water at larger quantities or with higher injection rates. Using highly diluted solutions may seem safer or be easier to pump, but it may not provide sufficient temperature generation to provide effective re-pressurization of the parent well.

For maximum re-pressurization, the goal should be to inject the most concentrated solutions that are practical and safe, thereby minimizing the amount of excess water that would otherwise absorb the reaction heat. If practiced correctly, this treatment technique can effectively modify the stress profile and redirect the fracture growth away from the parent well. The increased pressure in the parent well zone prevents hydraulic communication between the two wells and reduces the risk of fracture hits, thus preserving production performance. However, excessive heat and pressure during the reaction process may compromise well integrity or cause unintended fracture propagation outside the target zone. These risks could result in loss of zonal isolation, communication with unintended formations, or even wellbore damage. As a result, a detailed assessment is recommended to quantify the maximum temperature and pressure expected from the reaction under reservoir conditions, as well as the mechanical limits of the wellbore and completion hardware. Additionally, the potential for out-of-zone fracture growth, particularly in formations with stress anisotropy or weak bounding layers are present needs to be studied as well.

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